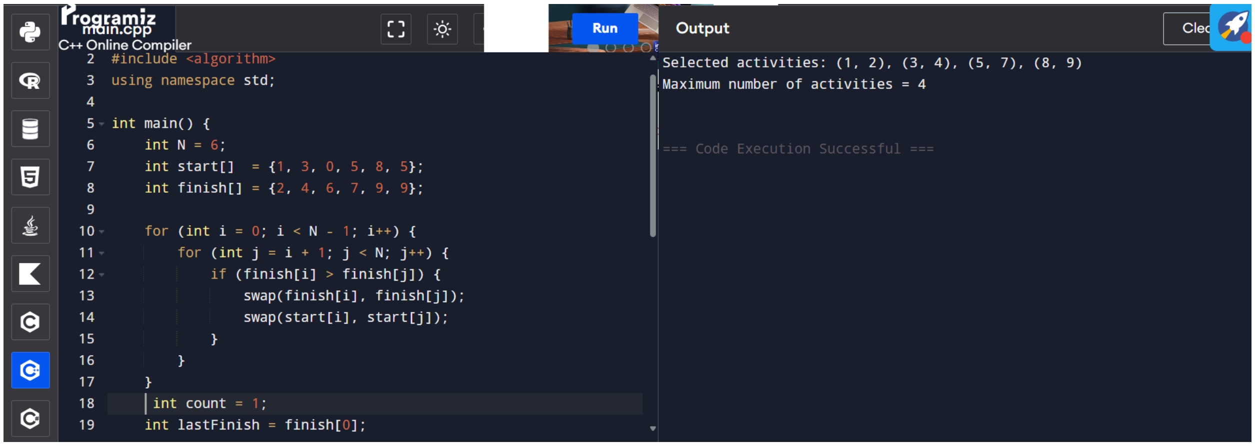
Ass_ 2_ DAA

Q1



```cpp
#include <algorithm>
using namespace std;

int main() {
    int N = 6;
    int start[]  = {1, 3, 0, 5, 8, 5};
    int finish[] = {2, 4, 6, 7, 9, 9};

    for (int i = 0; i < N - 1; i++) {
        for (int j = i + 1; j < N; j++) {
            if (finish[i] > finish[j]) {
                swap(finish[i], finish[j]);
                swap(start[i], start[j]);
            }
        }
    }
    int count = 1;
    int lastFinish = finish[0];
```

```
Selected activities: (1, 2), (3, 4), (5, 7), (8, 9)
Maximum number of activities = 4

=== Code Execution Successful ===
```

Q2

```cpp
    int arrival[]   = {900, 910, 920, 1100, 1120};
    int departure[] = {940, 1200, 950, 1130, 1140};

    sort(arrival, arrival + n);
    sort(departure, departure + n);

    int platforms = 1;
    int maxPlatforms = 1;
    int i = 1, j = 0;

    while (i < n && j < n) {
        if (arrival[i] <= departure[j]) {
            platforms++;
            i++;
        } else {
            platforms--;
            j++;
```

```
Minimum number of platforms required = 3

=== Code Execution Successful ===
```

Q3

```cpp
    int n = 3;
    Item items[] = {{100, 20}, {60, 10}, {120, 40}};
    int W = 50;

    sort(items, items + n, compare);

    double maxValue = 0.0;
    int capacity = W;

    for (int i = 0; i < n; i++) {
        if (items[i].weight <= capacity) {
            capacity -= items[i].weight;
            maxValue += items[i].value;
        } else {
            maxValue += items[i].value *
                        ((double)capacity / items[i].weight);
            break;
```

```
Maximum value = 220

=== Code Execution Successful ===
```

Q4

```
int main() {
    int n = 5;
    Job jobs[] = {
        {'1', 2, 100},
        {'2', 1, 19},
        {'3', 2, 27},
        {'4', 1, 25},
        {'5', 3, 15}
    };

    sort(jobs, jobs + n, compare);

    int maxDeadline = 3;
    int slot[3] = {0};
    char result[3];
    int totalProfit = 0;
    int count = 0;

```

```
Selected Jobs: J3 J1 J5
Maximum Profit = 142


=== Code Execution Successful ===
```

Q5

```
int main() {
    int n = 6;
    char chars[] = {'a', 'b', 'c', 'd', 'e', 'f'};
    int freq[]   = {5, 9, 12, 13, 16, 45};

    priority_queue<Node*, vector<Node*>, compare> pq;

    for (int i = 0; i < n; i++)
        pq.push(new Node(chars[i], freq[i]));

    while (pq.size() > 1) {
        Node* left = pq.top(); pq.pop();
        Node* right = pq.top(); pq.pop();

        Node* merged = new Node('#', left->freq + right->freq);
        merged->left = left;
        merged->right = right;

```

```
Huffman Codes:
f : 0
c : 100
d : 101
a : 1100
b : 1101
e : 111


=== Code Execution Successful ===
```